On the discussion questions regarding Prior's 'The runabout inference-ticket' and Belnap's 'Tonk, Plonk and Plink' (September 16, 2025)

Here are a couple of thoughts and observations on the discussion questions. Comments are welcome!

All groups:
"The key question behind the Prior and Belnap papers is this: 'What is a logical constant?' And behind that question is a broader one: 'Do all intuitively valid arguments depend (for their validity) on logical constants?' Discuss. What are the consequences of your answer for the scope of logic?"

The question 'What is a logical constant?' is in fact the question 'What counts as logic?'. That is a fundamental question but also a rather intractable one. So, rather than starting out with an attempt to answer that question, logicians prefer to 'lead by example'. They first present logical systems that everyone would agree capture, if not all of what logic is, then certainly its core parts. Then they explore various ways of changing and enriching these core systems with additional logical constants. The current course is no exception. It starts with proposition logic and first order predicate logic, two systems that everyone acknowledges as being logic. It also gestures towards extensions, such as tense logic and modal logic. It is interesting to note that there already opinions start to diverge. For example, according to some modal logic isn't strictly speaking logic, since it depends on concepts such as 'possibility' and 'necessity' that are, or so they would argue, not purely logical concepts.

This indicates that the question 'What is a logical constant?', or, equivalently, 'What is a logical concept?', can be ignored but cannot be made to go away. The Prior and Belnap papers focus on very basic features of 'logicality'. Prior's paper shows that admitting a concept/constant that makes everything follow from everything is self-defeating: it does away with logic altogether. Belnap then explores what constraint might rule out such concepts/constants and finds that conservativity does the job. That is an interesting move. But does it answer the question what a logical constant is? That depends. Is conservativity a necessary requirement, i.e., can there be a bona fide logical constant that is not conservative? Is it sufficient, i.e., are all logical constant that meet conservativity bona fide? Both, i.e., does conservativity *define* what a logical constant is?

One way to approach this issue is to start 'from the other side', so to speak, and ask whether what intuitively we find compelling forms of reasoning always need to be, or can be, accounted for in terms of expressions that count as logical constants. As a matter of fact, when we reason in an everyday context, we rely on a lot of implicit assumptions. 'If John left for Amsterdam this morning, he is not at home now', when said standing outside the door of his apartment in downtown Los Angeles at noon, is not something we normally would question. But it relies on a lot of assumptions: that people cannot be at two places at the same time; that it takes more than a couple of hours to fly from LAX to Schiphol Airport and back; that John not only left, but actually got on the plane; that the plane actually took off; that after take-off it wasn't made to return to LAX; that it wasn't diverted to a place close enough for John to be able to return home by now; and so on, and so on. Some of these assumptions are quite specific, other more general. What they have in common is that they are all *defeasible*:[1] 'If John left for Amsterdam this morning but wasn't allowed on the plane, he must be home again by now'. That is, we accept the

[1] Except perhaps for the 'not being at two places at the same time': that is hard to conceive as not holding. But does that make it 'logical' in any sense?

argument because we assume that the premise of the argument (the antecedent of the conditional) is true and that all other things are 'normal'.

Reasoning in real life is complicated and usually involves much more that purely formal considerations. Some draw the line right there and view these 'extras' as outside the realm of logic proper. But many logicians find it a challenge to develop logical systems that are capable of dealing with these additional factors. For example, they develop logics where it does not hold that if A implies B, then (*A* and C) also implies *B* (so-called 'strengthening the antecedent'), and thus account for one of the examples mentioned above. (See also the remarks on the question of group 1.)

Group 1:
"Belnap uses the Gentzen rules as a characterisation of deducibility (p. 132). Can you think of a piece or reasoning for which weakening does not hold? Or one that does not appear to satisfy permutation? If you can find such examples, what conclusions can be drawn from that?"

The rule of Weakening is what above was called 'strengthening the antecedent'. The rule states that if we can derive a conclusion *B* from premisses $A_1 \ldots A_n$ it also follows from any addition to those premisses: we can as it were 'weaken' the condition under which the conclusion follows by adding arbitrary premisses.[2] Logics that have Weakening (i.e., that allow 'strengthening the antecedent') are called 'monotone logics': if something follows from a set of premisses it will stay that way, not matter what additional we add to that set.

On many domains reasoning appears to be a monotone process, which makes this a justified property of a logical system that aims to capture that reasoning. Example: reasoning in mathematics is monotone. If can conclude that a certain theorem follows from a given set of axioms (and perhaps other already proven theorems), that conclusion does not change by adding another axiom (or bring in another theorem).[3]

But is reasoning always monotone? Arguably not. We already came across a possible exception in the remarks on the 'all groups' question. We often reason on the basis of incomplete information. We know some things and use them as premises in our arguments. But we hardly ever know everything there is to know about something. Waiting until we do have complete information is often not possible. But it is also not always necessary. In situations in which we do not know all the facts but are reasonably convinced that we do know all the relevant facts, we can reason and draw conclusions, acknowledging that additional information might defeat those conclusions but under the assumption that no such information will be forthcoming.

Group 2:
"Compare Prior's *tonk* to the argument that Lewis Caroll discusses in 'What the Tortoise said to Achilles' (1895) (paraphrased):

- Achilles : If *A* then *B* and *A*, then *B*
- Tortoise: Really? Why would that follow?

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[2] Note *B* is any arbitrary formula. It could also be '*not*-B', for example.
[3] Can we add anything? What if we add the negation of one of the premises? That does not matter: the result will be a set of premises that is contradictor, and from a contradiction anything can be derived. (This holds in most logics, para-consistent logics being the exception. We will come back to this at some point.)

- Achilles: Well, because if it is the case that if *A* then *B* and *A*, then *B*, and it is the case that if *A* then *B* and *A,* then it is the case that *B*
- Tortoise: Hm, and what motivates that?
- Achilles: Easy, that's because …

Discuss. What is wrong with an argument that is infinite in length?"

The main problem with arguments that are infinite in length is that they are not 'survey-able': you never know when they are done, so you can't judge whether they are any good. That is why in philosophy a common way of arguing against a particular analysis is by showing that it leads to an infinite regress: that each claim that support the analysis in its turn requires another claim to support it, which requires yet another claim, and so on. And analyses that lead to such a regress are considered principally flawed.

Note that this is not a problem of arguing *about* infinite 'objects', or about an infinite number of cases. Take the example of inductive proofs about the (potentially) infinite set of propositional formulas. In virtue of the finite character of the definition of that set, these proofs themselves are finite, even though they prove something about an infinite number of objects.

Infinite regress is the argument that Prior is referring to in the third paragraph of his paper: '…the old superstitious view that an expression must have some independently determined meaning before we can discover whether inferences involving it are valid or invalid.' This type of argument is effectively blocked when we are dealing with what are called 'analytically valid inferences'. For an inference is valid 'analytically' if it's validity follows just from the meanings of the words and phrases that are used. (Cf., the stock example 'All bachelors are unmarried', which is supposedly true simply because of what 'bachelor' and 'unmarried' *mean*.)

It is exactly this view on what constitutes validity that Prior targets. Nothing prevents us from stipulating the meaning of 'tonk' as Priori does. There is nothing wrong there, for a stipulation, after all, is just that: a stipulation. It is the idea that validity is a matter of things following from the meaning and the meaning only, that is the real culprit. In combination with 'tonk' and its meaning it leads to the consequence that everything follows from everything. And that is an absurd consequence. (For more on that see the remarks on the question of group 3.)

Group 3:
"Prior's argument can be seen as an instance of so-called '*Reductio ad Absurdum*' reasoning: you show that *A* is the case by showing that the assumption that *A* is not the case leads to an absurd conclusion (such as a contradiction). Identify this pattern in Prior's argument. Can you think of reasons (contexts, topics, …) in which *Reductio ad Absurdum* would be problematic?"

Although Prior himself does not refer to this form of reasoning, directly or indirectly, it seems pretty clear that the gist of his argument conforms to it. In the last paragraph his paper, Prior refers to the 'extreme convenience of the new form' of reasoning that 'tonk' makes possible: *A* follows from *B*, and *B* follows from *A*, for any *A* and *B*. And if anything follows from anything, there is no longer a need for logical 'hair-splitting' and all quarrelling about whether some argument is valid or not can be settled: they all are! This is a tongue-in-cheek way of pointing out that in fact adding 'tonk' to our logical repertoire leads to an absurd conclusion.

The target of this *reductio* argument is not 'tonk' as such. Rather it is the view that seems to make it a legitimate logical constant. That view is that logic is about 'analytically valid inferences', i.e.,

about inferences that hold purely on the basis of the meanings of the logical constants, meanings, moreover, that can be specified at will. (See also the remarks on the question of group 2.) Prior's introduction of 'tonk' exemplifies this: it defines 'tonk' by specifying what follows from '*A* tonk *B*' and from what '*A* tonk *B*' follows. And according to this view that is all there is to it. By showing that it allows for the introduction of 'tonk' and that the introduction of it leads to an absurd conclusion Prior thus means to show that the view that logic is about analytically valid inferences as such is untenable.

Group 4:
"A key assumption of Belnap's analysis is that one can characterise the role in deducibility of a connective independently from its meaning. Discuss. Can we analyse both completely independently? I.e., can you know what a connective means without any idea of what can be deduced from a sentence containing it? And reversely, can you know exactly what the role a connective in deductions is without knowing what it means?"

There is a sense in which the answer to these two questions is obvious: 'Yes, we can do that, because look: this is what we do!' We define the meaning of a connective by specifying a truth table, or by formulating a clause in the definition of valuations. And in doing so we do not need to refer to any valid or invalid inferences. And as we shall see later, when we discuss natural deduction, we can specify inference rules for connectives without mentioning their meaning.

So, what's the point of the question? The issue here is really, not what a formal system[4] *is*, but what we *use* it for. Consider the following definition of a language:

- vocabulary: {*a*,*b*}
- grammar:
  1. *ab* is a sentence;
  2. if *S* is a sentence, then *aSb* is a sentence;
  3. nothing is a sentence except on the basis of 1 and/or 2

This defines the context-free language that consists of all and only strings of the form $a^n b^n$, $n \geq 1$

This is a language, according to the definition that says that 'a language is a set of strings defined over an alphabet'. But what is it for? Is it a language that we can use to describe, analyse, communicate, …? Well, it could be, but then something more is needed than just this definition. We need to supply the language with a semantics that specifies how its expressions are connected, or with a set of rules for manipulating them that embody those connections. And the connections that account for in that way are the ones that are the ones that make our intended use of the language possible.

And this is in fact what we do when we define a logical system. Take propositional logic. We specify its semantics in such a way that we can *use* the logic to study valid and invalid inference patterns. Thus, the truth tables we use, or the definition of valuations functions we give, are geared to what we want to do with the system: analyse inference. That is not something that one can find *in* the system as such, it is a feature of how we define it and for what purpose. As we shall see, the same goes for when we define our logic using a system of inference rules. The rules define how expressions can be manipulated, the why of those manipulations resides in what we want to use the system for.

[4] In the broad sense: a logical system; a formalized theory of mathematics; a programming language; …

Thus, it is in the purpose(s) for which we have the system than the justification of its features resides. And whether that takes the form of the definition of a semantic interpretation or of the definition of inference does not matter, in the functionality of the system both come together. A lot more needs to be said about this, and we will come back to it when we study Ruy de Quiroz's paper.